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Weight of Evidence Method and Its Applications and Development

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Abstract

The development and applications about the weight of evidence technology in recent years are reviewed. This paper introduced the improved weight of evidence in remote sensing image processing and in different fields of application. Summary its constraints and existent problems. Look forward to the weight of evidence for the practical application.

Keywords: Weight of evidence; improved algorithm; applications; development status; prospects

1. Introduction

Weight of evidence is a geo-statistical method proposed by F.P.Agterberg, a mathematical geologist in Canada. It is based on binary images using a statistical model. Make composite analysis to some geological information, and so get the prediction of mineral potential areas. Each of these information is regarded as a evidence factor of the prediction of forming long-term. The contribution of each evidence factor is determined by its weight [1].

Weight of evidence originally was used in medical diagnostics [2], and later in the prediction of mineral ore field it was widely used. Now, its application spread other fields, such as landslide, oil and gas resources, fire, habitat and so on.

2. Weight of evidence and its theory

The core of weight of evidence method is to calculate the prior probability and posterior probability[3][4]. Take metallogenic prediction as an example. Assume that the study area is divided into T units in equal size, including D units for ore unit. For any evidence factor, its weight is defined as:

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$$W^{+} = \ln\left[\frac{P(B/D)}{P(B/\overline{D})}\right] \quad W^{-} = \ln\left[\frac{P(\overline{B}/D)}{P(\overline{B}/\overline{D})}\right] \tag{1}$$

In the equation, W^{+} and W^{-} respectively is the weight of existent evidence of factors area and inexistent area, and for area where loss original data, the weight value is zero. B is the cell number for the existent evidence of factors, and $\overline{B}$ is the cell number for the inexistent evidence of factors. $\overline{B} = T - B$. $\overline{D}$ is the cell number for non-mining unit. The conditional probabilities are as following:

$$P(B/D) = D \cap B/D$$

$$P(B/\overline{D}) = \overline{D} \cap B/\overline{D} \qquad P(\overline{B}/D) = D \cap \overline{B}/D$$

$$P(\overline{B}/\overline{D}) = \overline{D} \cap \overline{B}/\overline{D} \tag{2}$$

In the weight of evidence method, the requirement is that each evidence factor meets the conditions of independence relative to the mine point distribution. For n number evidence factors, if they satisfy above conditions, the possibility that any k units are the mining cell is the posterior probability , namely O. It is expressed using the logarithm.

$$\ln(O) = \ln\left(\frac{D}{T-D}\right) + \sum_{j-1}^{n} W_j^k \quad (j = 1,2,3,\cdots,n) \tag{3}$$

In the equation, W_j^k is the weight of j factor,

$$W_j^k = \begin{cases} W_j^{+} & \text{evidence factors j exist,} \\ W_j^{-} & \text{evidence factors j inexist,} \\ 0 & \text{original data loss.} \end{cases}$$

The posterior probability P = O / (1 + O). According to it, propose prediction.

Due to the both models are from conditional probability theory, the results are similar nature. However, later developed evidence weight models are more precise.

3. Implementation

The evidence weight method is an organic synthesis, including mathematical statistics, image analysis and artificial intelligence. By integrating spatial data from various sources, describing and analyzing the interaction between the data, and establishing the prediction model, so provide support for the decision makers. The method how to implement can be summarized as follows:

1) Collect the relevant information about impacting factors of the interested target. And carry on the screening, inducing, summarizing and digital processing. Then do the deal with the evidence factors. Because the evidence weight method is originally based on the binary raster image, the raster image should make binarize processing firstly, that is, translate all the topics into binary images. If using the vector information, it can be based on the grid model. First of all, extract and mesh the known interested point frame, then establish the grid layer with interested target.

2) Overlay the thematic layers of evidence factors and the interested target(grid) layer, then calculate the prior probability and weight (W +, W-).

3) Test the conditional independence of evidence factors relative to the interested target points. And according to the prior probability and the value of weight (W +, W-), identify the most reasonable thematic layers evidence factor. Lastly, calculate the posterior probability.

4) According to the results of posteriori probability, determine the forecasting scope of interested target.

4. Development and status of weight of evidence

4.1.Developed Algorithm

To evaluating the contribution rate of each factor on interested target factor, compare the data layer of interested target factor with various thematic layers, respectively [5]. To achieve this goal, the equation (1) is expressed in the form of the element number, such as the equation (4) below:

$$W_i^+ = \ln \frac{\dfrac{NpiX_1}{NpiX_1 + NpiX_2}}{\dfrac{NpiX_3}{NpiX_3 + NpiX_4}}$$

$$W_i^- = \ln \frac{\dfrac{NpiX_2}{NpiX_1 + NpiX_2}}{\dfrac{NpiX_4}{NpiX_3 + NpiX_4}} \qquad (4)$$

4.2.Prediction of mineral resources

In recent years, various data models are widely used in forecasting deposits and estimating resource, such as the weight of evidence model, fuzzy weights of evidence model, logistic regression model, fuzzy logic model, evidence of belief function model (EBF), neural network model and so on. There are advantages and disadvantages with each model[6] ~ [23].

Weight of evidence, as an important predictive model, in mineral resources prediction is successful. Geological Survey of Canada, based on weight of evidence in the Nova Scotia regions, did the research on GIS-based gold exploration and geological data, and the evaluation of mineral resources. Develop the Comprehensive Assessment System of grid-based data structure for GIS multi-source information. China Academy of Geological Sciences on the MAPGIS software platform developed the Mineral Resources Assessment System (MRAS), which also includes the weights of evidence predictive model. Anhui Geology and Mineral Resources Bureau used weight of evidence to do the research on gold resources evaluation in east.

4.3.Oil and Gas Resource Assessment

Weight of evidence model, which is on Bayesian conditional probability, is a common method of oil and gas resource evaluation on the platform of GIS. First of all, analyze the conditions on the natural gas reservoir. Then establish source rocks layer, seals layer, reservoir layer, fault layer, pressure layer and other geological evidence layer in the system. Secondly, use the overlay analysis capabilities of weights of evidence model to calculate the layer parameters of each evidence layer, and test the independence. Finally, generate the posterior probability map of the potential exploration of gas resources. According to the relative size of the posterior probability, divide the forecast area into Ⅳ levels. Stipulate the district above Ⅱ level as a favorable target gas accumulation. All these provide decision support for further exploration [24] [25].

4.4.Zone landslide hazard

Weight of evidence model was applied to landslide hazard zoning, whose point objects are landslide points and the evidence layer is the geological and thematic maps which is favorable for landslide [26].

Do the research on landslide hazard zoning on the Yangtze River Three Gorges Zigui - East Pakistan by the weight of evidence method. Data sources mainly include geological maps, SPOT5 multi-spectral satellite image data and digital elevation model (DEM) and so on. Make use of the function of data extraction and analysis in RS and GIS, to extract the geology (lithology and structure), topography (slope, aspect, water, elevation, ravines buffer, density), hydrogeology (vegetation index) and the destruction of power (the information about influential factors in the study area, such as river buffers). Grade above influential factors to establish several evidential layer. Then on the basis of the distribution of known landslides in the different layers, determine the weights of appropriate evidence layer. Finally, according to the superposition of the weight value of different factors, make sure of the probability of a landslide unit. The results are in good agreement compared with the existing distribution of landslides. In this way, it can evaluate objectively the extent of various factors for the development of landslide. Then select the assessment factor of landslide risk (evidence factor) and evaluate the degree of risk quantitatively.

Observational information and experience have shown that the probability of landslide occurrence depends on the internal variables and external variables working together. However, the external variables vary of the places, and distribute over time. Father more, because lack of information about the spatial distribution, it is difficult to be estimated. Therefore, in the practice of landslide hazard assessment, landslide susceptibility mapping has not considered the external variables of landslide event.

Although the weight of evidence has not been applied to map landslide susceptibility, because some studies have successfully applying this technique and some survey on the characteristics of spatial relationships and distribution, so the weight of evidence is obvious applicability in a small watershed for map landslide susceptibility.

4.5.Estimate fire possibility

Recently, weight of evidence has been used to predict forest fire [27] [28]. The reason why select this method is as the following two. Firstly, it can directly display the spatial relationships between the fire occurred and the data of tables and maps. Secondly, it’s not necessary for normal school assumption of data distribution.

Taking DaXinAnLin Huzhong forest as the study area, consider the natural and human factors which may affect the occurrence of forest fire. Then, select the factors that can greatly reflect the topography, weather, fuel and human activities from those numerous factors. Analyze the possibility of forest fire by the weight of evidence. The results showed that there are some uncertainties of forest fires, but the research still has a good guide for the management of forest in the region. It provides the scientific basis for the choice of forest fire risk zone, priority areas of forest fuel. It should strengthen the foundation research of forest fuel, especially probe deeply into the spatial and temporal distribution of the forest fuel, and improve the accuracy of spatial data. So reduce uncertainty, and thus make scientific predictions for forest fires better.

4.6.Adaptability of animal habitat

It is a new field of weight of evidence method used in animal habitat adaptability research in recent years [29]. In view of rationality, this work analyzes the spatial existing data sets to carry out rapid assessment of habitat quality. In order to improve the management plan of existing wetland, there provide an objective evaluation of habitat suitability in different aspects. Using of indicator species and the prediction model, it can identify the causal relationship during species distribution, environmental data and ecological condition, and offer the distribution of rare species and assessment of suitable areas. All are challenge.

Evaluate the habitat adaptability of red-crowned crane by weight of evidence in Zha Long. Using spatial analysis function in RS and GIS, extract land use, vegetation, railways, national highways and residential information, and establish thematic layers of evidence, then determine the appropriate weight value of the evidence layer, test the conditional independence, and finally generate the posterior probability map. It is an objective quantitative evaluation of weight of evidence method for habitat adaptability. Similarly, an objective evaluation of habitat suitability model could assess the value of biological diversity, manage and protect of wetlands, provide intellectual support for protecting valuable habitat policy.

5. Problems

In recent years, both at home and abroad the research on weight of evidence is gradually in depth. Weight of evidence method is a data-driven approach, and it is easy to program. In the face of actual complex environment system, this algorithm has begun to attach importance because of its unique ability to deal with problems in many areas, and becomes a great potential algorithm. However, weight of evidence algorithm still needs further improvement, and in practice there are many problems need to be studied in different fields, such as:

(1) The weight of evidence used for evaluation of mineral resources and application go to mature, but for other applications fields is tentative. This is decided by the established conditions of weight of evidence model, and these are factors to be considered. The improvement can make up for the algorithm, making the prediction is closer to the actual situation and more valuable.

(2) The weight of evidence is data-driven method. It has the advantage that data-driven avoids the subjectivity of weight to choose, it is easy to program and compose multiple maps. However, the disadvantage is the map for the analysis must be independent, and the target layer must include the known interested points. So, it is not suitable for the interested target less areas. In the addition, if this method is equivalent to lack of data and adverse conditions, but also interested in target layer containing known and therefore not suitable for low level of interest in target research areas used. On the other hand, if weight of evidence is used in above addition, a feasible method is that add the auxiliary feature layers comprised of possible forecasting types in the study area, the expert opinions, a variety of spatial data sources and other mature conditions.

(3) The weight of evidence has developed rapidly in the forecasting mineral resources. It should be noted that the success of the use of GIS to predict mineral resources is still largely depends on the level of awareness for the law of mineralization and reasonable forecasting methods. So in other fields, it develops still in infancy, which to some extent depends on the forming law of the interested target in study area and the rationality applications of the prediction model.

(4)The weights of the layer evidence, to some extent, directly impact on the accuracy of the method. How reasonably determine the layer weight, which is worthy of further study.

6. Summary and outlool

The applications of weight of evidence mostly focused on prediction of potential mineral resources. It is a causal link method of quick identification between species distribution and environmental data. In identifying, understanding and prediction of quantitative analysis of evidence factors and the links species, the weight of evidence is an effective tool. The method is objective, especially in the choice of weighting factors, which is more easily explained than the regression coefficients method. The model describes and analyzes the interaction of spatial data, and provides intellectual support for the policy makers, which is an effective tool.

Weight of evidence combined with spatial data to describe and analyze the interaction, at the same time, which provides decision support for policy makers and generates a prediction model. There will be a greater contribution in resource utilization, disaster prediction, environmental protection and other aspects.

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